

Mahdi Saleh

Munich, Germany

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WORK EXPERIENCE

Senior Applied Scientist Oct 2024 - Present

Zillow Inc, Germany

Working on 3D computer vision algorithms to enable rich 2D and 3D experiences.

Senior Research Engineer

July 2024 - Sept 2024

Virtual Staging AI (Acquired By Zillow Group)

Working on developing an instant furniture staging for real estate photography.

Research Associate

Sep 2019 - June 2024

Technical University of Munich, Munich

- National AI-Safeguarding Project: 3D object detection and 6DoF pose estimation from 3D data and 3D human pose estimation from RGB camera.
- Researcher on point cloud and mesh processing, 3d pose estimation, graph neural networks, hierarchical understanding, and self-supervised learning.

PhD Research Scientist Intern

Jul 2021 - Aug 2021

Facebook Reality Labs, Sausalito, CA

Research Engineer

Nov 2017 - Aug 2019

Framos GmbH, Munich

- National IRIS Project: Autonomous robotic container unloading using 3D perception and machine learning
- Leading several projects on reinforcement learning, human pose estimation, reconstruction, and depth estimation.

Augmented Reality Developer

Jul 2017 - Nov 2017

AR-Experts, Munich

- Design and implementation of augmented reality training apps for Microsoft HoloLens

Artificial Intelligence Engineer

May 2017 - Jul 2017

IBM Watson IoT Center, Munich

- Design and implementation of a wearable helping low vision people in product recognition in supermarkets using deep object recognition and gesture control

Research Developer

Aug 2015 - Mar 2017

Framos GmbH, Munich

- Extracting anthropometric data of humans using contour alignment and marker detection
- Pallet dimensioning and 3D reconstruction for logistic pipelines

EDUCATION

Doctor of Philosophy (Ph.D.) - Computer Science

Jan 2019 - Jan 2025

*Technical University of Munich, Munich - passed with highest distinction, **summa cum laude***

Thesis title: Hierarchical Representations for 3D Shape Understanding

M.Sc. of Biomedical Computing - Computer Science
Technical University of Munich, Munich

Oct 2014 - Oct 2017

B.Sc. of Electrical Engineering
Iran University of Science and Technology

Sep 2008 - Sep 2013

SKILLS

3D computer vision, sparse view reconstruction, Geometry processing, Deep learning, 3D Detection, and 6D Pose Estimation, Graph Neural Networks, PyTorch, Computer Graphics, Self-supervised Learning, Generative Media, 3D Gaussian Splatting and novel view synthesis

HONORS & AWARDS

CVPR Outstanding Reviewer	2021
Zeidler-Mint Makerspace Scholarship Winner	2016
TUM HackRoboy 1st place winner	2016
Elite Student of among the whole class of electrical engineering	2008

SELECT PUBLICATIONS

CVPR 2026, ConceptPose: Training-Free Zero-Shot Object Pose Estimation using Concept Vectors, L. Kuang, Y. Velikova, M. Saleh, J. Zaeche, DP Paudel, B. Busam

CVPR 2025, ESCAPE: Equivariant Shape Completion via Anchor Point Encoding, B. Bekci, N. Navab, F. Tombari, M. Saleh

CVPR 2022, *Bending Graphs: Hierarchical Shape Matching using Gated Optimal Transport*, M. Saleh, SC. Wu, L. Cosmo, N. Navab, B. Busam, F. Tombari

CVPR 2022, *ZebraPose: Coarse to Fine Surface Encoding for 6DoF Object Pose Estimation*, M. Saleh, Y. Su, T. Fetzner, J. Rambach, B. Busam, N. Navab, D. Stricker, F. Tombari

Neurips 2021, *CoFiNet: Reliable Coarse-to-fine Correspondences for Robust PointCloud Registration*, H. Yu, F. Li, M. Saleh, B. Busam, S. Ilic

3DV Oral 2020, *Graphite: GRAPH-Induced feature Extraction for Point Cloud Registration*, M. Saleh, S. Dehghani, B. Busam, N. Navab, F. Tombari

LEADERSHIP EXPERIENCE

BMC Master Program Coordinator TUM	2020 - 2022
Head of IEEE student branch IUST	2011 - 2012
Vice President of Scientific Committee of EE faculty at IUST	2010 - 2011

LANGUAGE SKILLS

English: Fluent (C1), German: Intermediate (B2), Persian: Native